

Bioinspired Sulfobetaine Borneol Fluorinated Amphiphilic Polymers for Marine Antifouling and Fouling Release Applications

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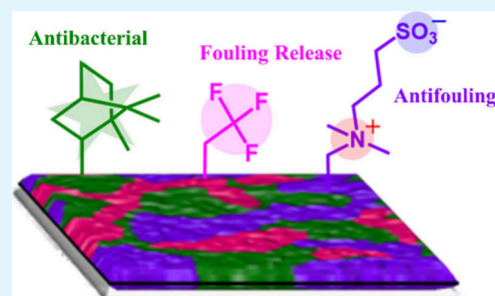
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ABSTRACT: The development of nontoxic antifouling coatings in static marine environments is urgent. Herein, the successful synthesis of sulfobetaine borneol fluorinated polymers (PEASBF) by a free radical polymerization method is reported. The PEASBF coatings exhibit outstanding antifouling activity, which effectively resists the adhesion of *Bovine serum albumin* (FITC-BSA adhesion rate: 0.5%), *Pseudomonas sp.* (Biofilm: 1.3 absorbance) and *Navicula sp.* (Diatom attachment rate: 33%). More importantly, the PEASBF coatings display outstanding fouling release properties, achieving a release rate of 98% for *Navicula sp.*, and the absorbance of the *Pseudomonas sp.* biofilm is only 0.2 under 10 Pa shear stress. XPS and MD studies showed that the fluorinated/isobornyl groups induce more sulfobetaine groups to migrate toward polymer surfaces for intensify antifouling. Additionally, the chiral stereochemical structure of borneol enhances antifouling and fouling release ability of amphiphilic polymers. Therefore, the PEASBF has the potential for static marine antifouling applications.

KEYWORDS: biofouling, amphiphilic, isobornyl methacrylate, fluorine, zwitterionic



1. INTRODUCTION

Microorganisms, algae, and other organisms grow and settle on the underwater surfaces of ships, pipes, and fishing nets, forming marine colonies. This marine biofouling phenomenon has the potential to decrease ship speed, increase roughness, elevate fuel consumption, cause economic losses, and facilitate biological invasion.¹ Due to these adverse factors, there has been a gradual emergence of antifouling (AF) and fouling release (FR) coatings, which either prevent biological sedimentation or facilitate the detachment of marine microfouling and larger fouling from surfaces.² In recent years, amphiphilic coatings have become research frontiers and hot topics due to the combination of AF and FR properties in static marine environments.³ Amphiphilic materials are designed and inspired by the microphase separation of the inner wall surface structure of blood vessels, that can effectively prevent protein adhesion.^{4–6} Although the exact mechanism is still being investigated, the associated hydration of hydrophilic groups is beneficial in resisting biological adhesion, and the combination between hydrophilic components and hydrophobic components can enhance fouling release properties of coatings.^{7–9}

In the field of amphiphilic polymer research, hydrophobic polymers are typically selected from a range of options including butyl methacrylate, fluorinated polymers, PDMS, etc.^{10–14} Among these polymers, fluorinated polymers possess superior properties such as high thermal stability, excellent chemical inertness, and low surface energy.¹⁵ In fluorinated

polymers, the fluorine element is easy to accumulate on the surface, which greatly reduces their surface free energy. This feature confers resistance to the rearrangement and penetration of marine adhesions, which can be dislodged in the washing or shaking action of seawater.¹⁶ However, some certain marine organisms, such as *Navicula perminuta*, adheres more readily to fluorinated polymer surfaces.¹⁷ Due to such a disadvantage, numerous researchers have chosen to employ polyethylene glycol as a hydrophilic compound, aiming to mitigate the adhesion of *Navicula perminuta*.^{10,11,13} Since polyethylene glycol materials are prone to auto-oxidize (forming aldehydes and ethers) in the presence of oxygen, the zwitterionic materials have drawn increasing attention in marine antifouling applications due to the formation of a strong hydration layer through electrostatically induced hydration or ionic solvation, which can efficiently resist nonspecific protein adsorption.¹⁸ Zwitterionic functionalized surfaces are able to effectively inhibit marine micro- and macro-organisms, such as bacteria,¹⁹ diatoms,⁷ barnacles,¹² and mussels.²⁰ Therefore, zwitterionic compounds are commonly used to improve the AF and FR properties of hydrophobic polymers. Unfortunately, the

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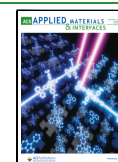


Table 1. Characterization Data of Polymers

Samples	MMA/EA/SBMA/IBOMA/TFEMA ^a	MMA/EA/SBMA/IBOMA/TFEMA ^b	$M_n \times 10^{-3}$ (g/mol) ^c	$M_w \times 10^{-3}$ (g/mol) ^c	PDI ^c	T_g (°C) ^d
PEA	26.7/13.3/0/0/0	26.4/13.6/0/0/0	64	90	1.4	30
PEAF	24.0/12.0/0/0/4.0	23.1/11.9/0/0/5.0	47	62	1.3	25
PEAS	24.0/12.0/4.0/0/0	23.4/12.0/4.6/0/0	63	89	1.4	25
PEASF	21.3/10.7/4.0/0/4.0	21.0/11.1/4.7/0/3.2	60	82	1.4	28
PEASBF	16.0/8.0/4.0/8.0/4.0	15.4/7.9/4.6/8.3/3.8	58	81	1.4	38

^aFeed weight ratio (g). ^bDetermined by ¹H NMR (g). ^cDetermined by GPC. ^dDetermined by DSC (°C).

synthesis of amphiphilic polymers via combining zwitterionic groups with fluorinated polymers is challenging due to the incompatibility of zwitterionic compounds and fluorine materials.

The antifouling properties of amphiphilic polymers can be further enhanced by repelling or killing contact bacteria. According to the principle of environmental friendliness, natural or biomimetic strategies are adopted as much as possible to repel marine bacteria adhesion.²¹ Borneol, a natural compound with unique chiral characteristics and excellent antibacterial adhesion properties, has become a potential compound with antifouling ability.²² The isobornyl methacrylate (IBOMA), with borneol antibacterial structures, exhibits good bacterial adhesion resistance, high rigidity, and desired thermal stability.^{23,24} The antibacterial mechanism of polymers with borneol structures mainly affects the surface recognition of bacteria on the surface due to its chiral stereochemical structure, thereby preventing bacterial adhesion.²⁵ It has been reported that the IBOMA can enhance the antifouling properties of hydrophobic and hydrophilic polymers.²⁶ For example, Song et al. revealed that borneol fluorinated polymer is able to effectively prevent antibacterial and antifouling activity.²¹ Cheng et al. prepared an antifouling and antibacterial hydrophilic poly[(2-methacryloyloxyethyl phosphorylcholine)-st-(4-formylphenyl methacrylate)-st-(isobornyl acrylate)] coatings which can prevent protein adsorption and inhibit bacterial adhesion relying on the hydrophilic surface and the special stereochemical structure.²⁵ These studies provided valuable insights that IBOMA has the capability to copolymerize with zwitterionic and fluorine monomers, potentially alleviating the incompatibility between zwitterionic and fluorine groups, thereby enhancing their antimicrobial activity.

In this work, according to the principle of environmental friendliness, the IBOMA as a bioderived antibacterial module is used to enhance the antifouling ability of amphiphilic polymers consisting of fluorine as a hydrophobic building block and sulfobetaine as a hydrophilic building block, which aims at prevention of microfouling in static marine environments. The designed novel bioinspired antibacterial amphiphilic polymer (PEASBF) is prepared by a free radical polymerization with 2,2,2-trifluoroethanol and 2-acetoxy-1-methoxypropane as solvents. The effects of functional monomers on their antibacterial properties, antifouling properties, and fouling release properties, are studied, and the mechanism is further analyzed. This research provides more possibilities for amphiphilic antifouling coatings for static marine applications.

2. EXPERIMENTAL SECTION

2.1. Materials. 2,2,2-Trifluoroethanol (TFE) and 2-acetoxy-1-methoxypropane (PGMEA) were provided by Anhui Energy Chemical Co., Ltd. 2,2,2-Trifluoroethyl methacrylate (TFEMA), [2-

(methacryloyloxy)ethyl]dimethyl-(3-sulfopropyl) (SBMA), isobornyl methacrylate (IBOMA), dibenzoyl peroxide (BPO), and 2,2'-azobis(2-methylpropionitrile) (AIBN) were supplied by Shanghai Aladdin Bio-Chem Technology Co., Ltd. Methyl methacrylate (MMA) and ethyl acrylate (EA) were obtained from Xilong Scientific Co., Ltd. Artificial seawater (ASW) was obtained following ASTM D1141-98. All reagents were used as received.

2.2. Synthesis of Sulfobetaine Borneol Fluorinated Polymers (PEASBF). First, 4 g of PGMEA and 16 g of TFE were added to a four-neck flask equipped with a cooling tube, thermometer, and dropping funnel. The solvent mixture was then heated to 95 °C. Next, 16 g of MMA, 8 g of EA, 4 g of TFEMA, 8 g of IBOMA, 4 g of SBMA, and 1 g of AIBN were added to a mixed solvent consisting of 10 g of PGMEA and 40 g of TFE. The above-mixed solution was added dropwise into a four-necked flask at 95 °C for two h under the protection of a nitrogen atmosphere. After the addition was completed, 0.2 g of BPO was dissolved in 10 g of PGMEA, and then, the mixture was added to the four-necked flask. The temperature was maintained at 95 °C for four h, and then the mixture was cooled to room temperature to obtain the PEASBF polymer. In the above synthesis process, the PEASBF polymer was prepared via radical polymerization initiated by AIBN and BPO. AIBN was dissolved with the monomers and initiated polymerization in the first stage, and a relatively high amount of AIBN (2.5 wt %) was beneficial for the removal of polymerization inhibitors from the system. BPO was added in the second stage to increase the monomer conversion degree as chaser.²⁷ The synthesis process of acrylate polymers (PEA), fluorinated polymers (PEAF), sulfobetaine polymers (PEAS), and sulfobetaine fluorinated polymers (PEASF) was the same as that of PEASBF. The raw material composition of the polymers is shown in Table 1. As a result of the absence of volatile matter utilization or production, the mass ratio of the monomers in the polymer was almost the same as the feed weight ratio.

2.3. Characterization. ¹H NMR spectra were recorded on an NMR spectrometer (Bruker AVANCE III 500) using CDCl₃ as a solvent and tetramethylsilane (TMS) as an internal standard. ATR-FTIR spectra were recorded on a Nicolet 6700 infrared spectrometer in the 4000 to 400 cm⁻¹ wavenumber range. The number-average molar mass (M_n), the mass-average molar mass (M_w) and the polymer dispersity index (PDI) of polymers were performed using an Agilent LC1200, USA, with DMF as the mobile phase, at a temperature of 25 °C, a flow rate of 1 mL/min, and a sample concentration of 1 mg/mL for gel permeation chromatography (GPC). The 3D surface profilers (NewView9000, ZYGO Corporation) were used to test the surface roughness and topography. The surface structure was examined by a multifunctional X-ray photoelectron spectroscopy instrument (Thermo ESCALAB 250Xi). The values of water contact angle were obtained on a contact angle system (KRÜSS DSA25S, Germany) with 4 μL of deionized water dropped at 25 ± 1 °C, and the values fitted by Young–Laplace were taken from at least five different measured positions on the polymer surface. A pull-off test measured the adhesion of the coating to the epoxy substrate.

2.4. Thermal Analysis. Differential scanning calorimetry (DSC) was tested on a differential scanning calorimeter (DSC2500, USA). Samples were first heated to 150 °C and kept for 5 min, then cooled to -20 °C and finally reheated to 150 °C. The rise/fall rate at each stage was 10 °C/min. Thermogravimetric analysis (TGA) was tested on a thermal gravimetric analyzer (Q500, USA). Samples were heated

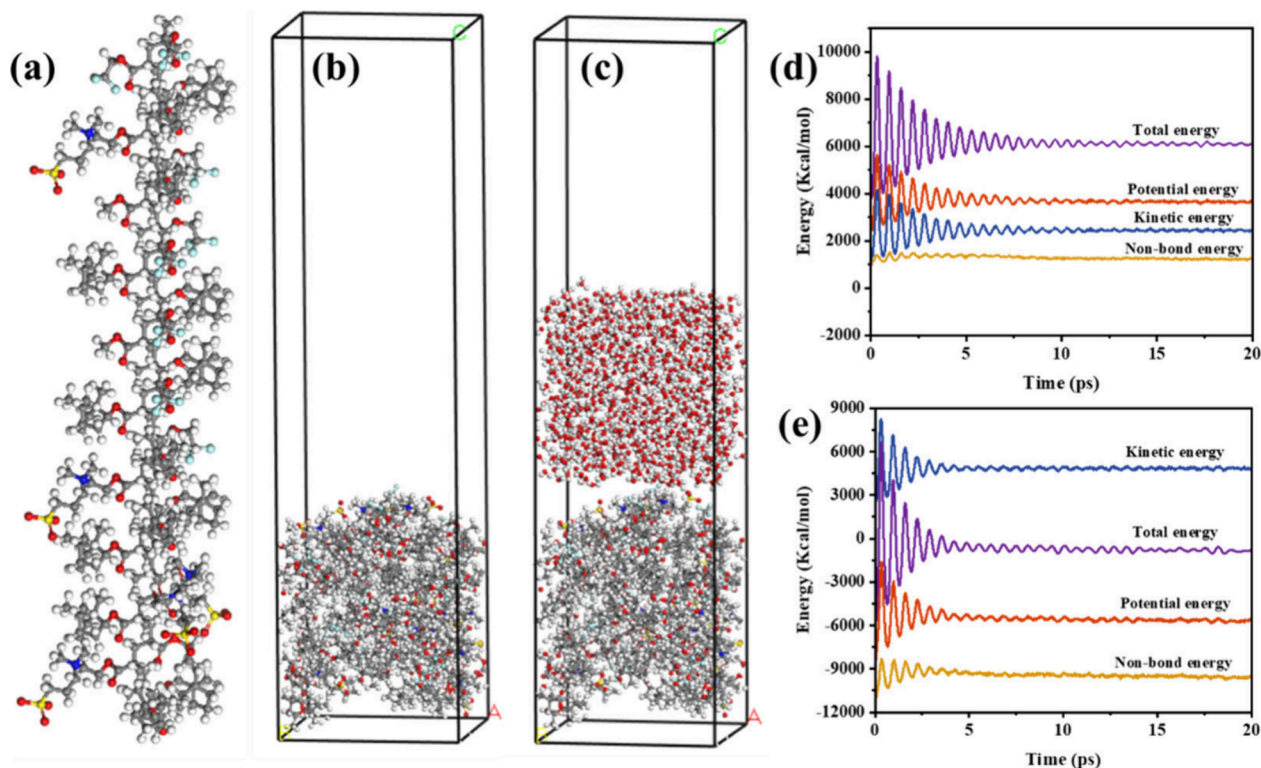


Figure 1. (a) Optimized structural model of PEASBF. A representative snapshot of the initial model of PEASBF (b) in a vacuum and (c) in the presence of water. The energies evolve with time (d) in vacuum and (e) water environments.

to 800 °C from 25 °C at a heating rate of 10 °C/min (N₂ atmosphere).

2.5. Water Absorption. The glass slides coated with polymers (20 × 20 mm²) were immersed in artificial seawater and taken out after immersion in the water for 7 days. The liquid on the surface of the samples was swabbed with lint-free paper. The following equation calculated the water absorption:

$$\text{Water absorption rate} = \frac{W_1 - W_0}{W_0}$$

Where W_1 in the formula represents the mass of the polymers after immersion in the artificial seawater for 7 days, and W_0 represents the original mass of the polymers.

2.6. Static Protein Adhesion Tests. The fluorescein isothiocyanate labeled bovine serum albumin solution (FITC-BSA, 100 μL) was diluted in 20 mL of PBS solution, and the 15 mg/L of FITC-BSA solution was prepared according to the literature method.²⁸ The glass slides coated with polymers were put into 10 mL of a FITC-BSA solution. After standing for 24 h in the dark room, the samples were washed with PBS solution for 1 min under ~0.2 Pa shear stress to remove unattached FITC-BSA. The shear stress was determined by a rotating disk system prepared according to the literature.²⁹ Then, the attached FITC-BSA on sample surfaces was observed by an inverted fluorescence microscope (Olympus IX71, Japan), and the FITC-BSA adhesion rate was calculated by ImageJ.

2.7. Bacterial Adhesion and Release Tests. The antibacterial activity of polymers was investigated by using *Pseudomonas sp.* as a marine fouling bacteria. The *Pseudomonas sp.* was cultured in the beef-protein medium at 37 °C. Place 10 mL of *Pseudomonas sp.* suspension (OD₆₀₀ = 0.015) was added in cell culture plates with polymer samples (20 × 20 mm²). The growth curve of the *Pseudomonas sp.* was obtained by measuring the absorbance of bacteria suspension at 600 nm every 2 h. The samples were immersed in a bacterial suspension for 12 h and washed with PBS solution. The samples were stained by LIVE/DEAD Bac Light reagent.³⁰ The stained samples were visualized by an inverted fluorescence microscope (Olympus IX71, Japan) to obtain the adhesion of bacterial cells. After incubating for 24

h, the samples were washed with PBS solution for 1 min under ~0.2 Pa shear stress to remove unattached bacteria cells.²⁹ The biofilm attached to the coating was stained with crystal violet (0.1 wt %), and the excess crystal violet solution was washed with PBS solution. After taking pictures of the stained samples, the crystal violet was extracted with aqueous acetic acid (10 vol %) and was measured at 590 nm to obtain the absorption intensity. The bacterial release test was performed with 1 or 10 Pa shear stress for 1 min. The rest of the photographing and extraction of the crystalline violet process was the same as above. The shear stress was determined by a rotating disk system prepared according to the literature.²⁹

2.8. Diatom Adhesion and Release Tests. The diatom adhesion and release tests of polymers were researched by using *Navicula sp.*. The *Navicula sp.* were subjected to a 12 h/12 h light-dark cycle at 25 °C in an F/2 medium. Place 10 mL *Navicula sp.* suspension (OD₄₄₀ = 0.015) was added in cell culture plates with polymer samples (20 × 20 mm²). After incubating for 24 h, the samples were washed with ASW for 1 min under ~0.2 Pa shear stress to remove unattached diatom.²⁹ Then, the samples were immediately placed under an optical microscope to take pictures, and 15 randomized fields of view were selected for quantitative counting. The diatom attachment rate was calculated as the diatom density of the coated surface divided by the maximum diatom density. The diatom release test was performed with 1 or 10 Pa shear stress for 1 min. The rest of the photographing process was the same as that above. The following equation calculated the diatom removal rate:

$$\text{Diatom removal rate} = \frac{D_0 - D_1}{D_0}$$

Where D_1 in the formula represents the diatom density of the coated surface and D_0 represents the maximum diatom density.

2.9. Molecular Dynamics (MD) Simulations. The PEA, PEAF, PEAS, PEASF, and PEASBF polymer molecules were constructed using the Materials Studio, and each polymer molecule was built with 30 repeating monomer molecule units (Figure 1a). The molar ratio of monomers in polymers were determined based on experimental information. Noteworthy, the molar ratio of functional monomers in

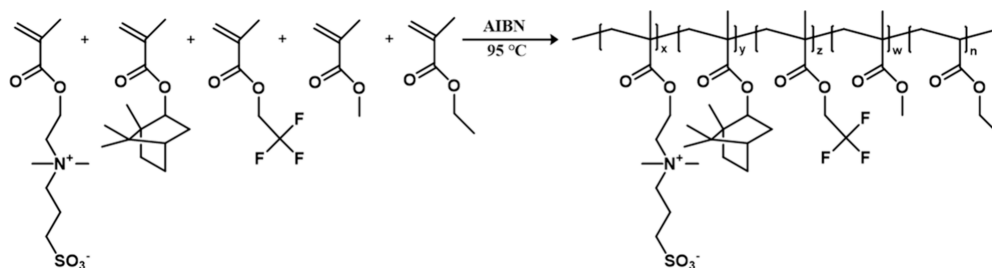


Figure 2. Synthetic route of the PEASBF.

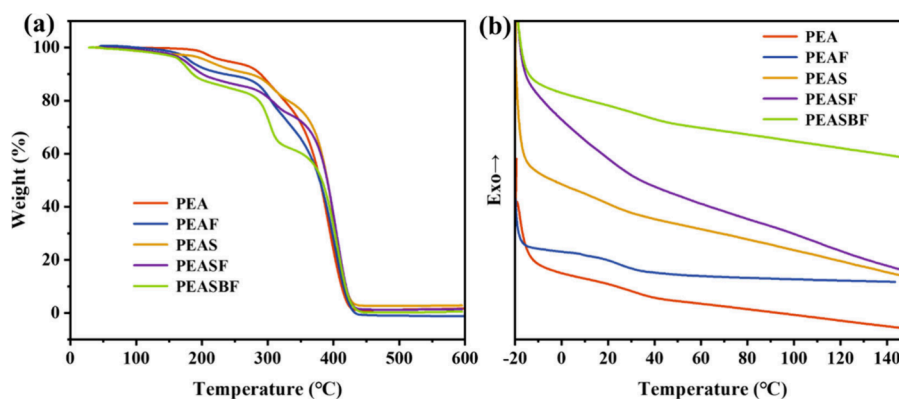


Figure 3. (a) TGA and (b) DSC curves of the polymers.

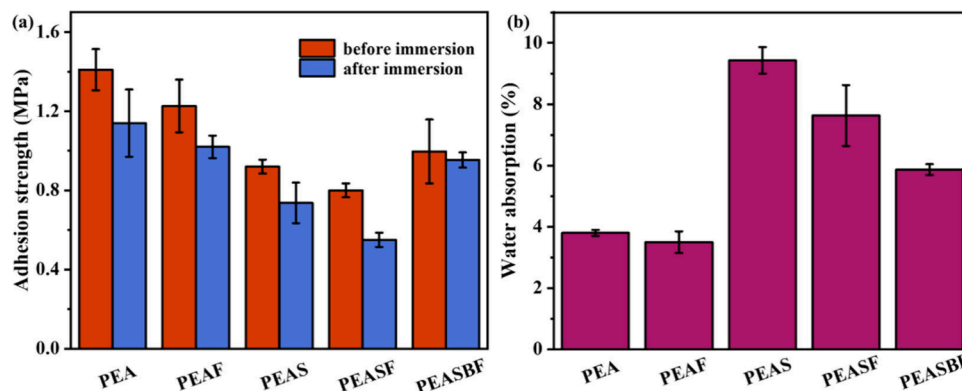


Figure 4. (a) The adhesion strength of polymers to the epoxy board. (b) The water absorption of polymers after immersion in the artificial seawater for 7 days.

MD simulation was four times than that in experimental, which is conducive to more useful information. The molar ratio in MD simulation, degree of polymerization, total number of atoms, and box size of equilibrated structures is reported in Table S1.

The polymer models with a density of 1.19 g/cm³ were constructed by an “Amorphous Cell” module of Materials Studio, and each of the models was subjected to five repeated annealing cycles under NVT (constant-volume and constant-condition) conditions starting from 300 to 500 K to achieve reliable energetically favorable states molecular conformations. Then, the water models (30.0 × 30.0 × 30.0 Å³) with the density of 1 g/cm³ and the TFE models (30.0 × 30.0 × 30.0 Å³) with the density of 1.39 g/cm³ were constructed by the same way. The constructed models were subjected to dynamics simulations under NVT conditions at 298 K to obtain a uniformly distributed system. In order to calculate effect of hydration on polymers, the water layer was placed between the polymer surface and the vacuum layer (thickness = 40 Å), and a representative snapshot of initial model of PEASBF in the presence of water was shown in Figure 1. The established systems were calculated with time steps of 1.0 fs in the NVT condition at 298 K, and equilibration was carried out for 20

ps. The energy curves were stable and no strong fluctuation occurred after 5 ps simulations, indicating that the system achieved an equilibrium (Figure 1d-e), and the last 5 ps trajectory was used for analysis. The binding energy between polymers and TFE was calculated by the same way.

2.10. Marine Field Test. The Marine field test was performed at Dalian Harbour, Dalian (38°N, 121°E) in China from February 2024 to June 2024. The epoxy resin panels coated with synthesizing polymers were immersed in seawater at depths of 0.2–2.0 m. After immersion in seawater for a certain period, the coated panels were taken out of the seawater and photographed to record the attachment of fouling organisms.

3. RESULTS AND DISCUSSION

3.1. Structure and Characteristics of the PEASBF.

Figure 2 presents the synthetic route of the PEASBF, which was synthesized with MMA, EA, TFEMA, IBOMA, and SBMA monomers via free radical polymerization. Tables 1 lists the feed weight ratios and molecular weights of the synthesized

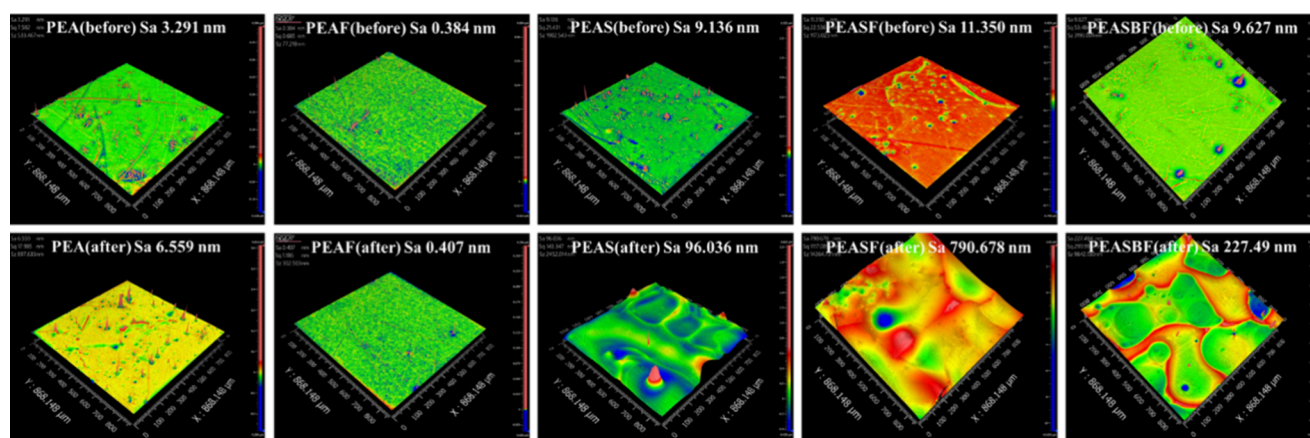


Figure 5. Typical surface morphology and corresponding roughness of polymers before and after immersion in artificial seawater for 7 days.

polymers. The successful synthesis of PEA, PEAf, PEAS, PEASf, and PEASBF was confirmed via ATR-FTIR and ^1H NMR (Figures S1, Supporting Information). Detailed analysis of these characterizations can be found in the Supporting Information.

The thermal stabilities of the as-obtained polymers are shown in Figure 3. Specifically, there were three distinct weightlessness stages in the TGA curve of PEASBF. The first stage was observed at 150–200 °C, which can be ascribed to the evaporation of residual solvents. Compared to other polymers, a significant weight loss at 270–320 °C is observed in the TGA curve of PEASBF, which is attributed to the loss of the side chains by the cleavage of IBOMA monomers, resulting in the release of camphene as a product.^{31,32} At this weightlessness stage, a similar pattern is seen in PEA without IBOMA monomers, attributed to the decomposition of small molecule polymers.²⁸ Additionally, the weight loss at 370–420 °C is the scission reaction of the C–C bond on the main chain, releasing monomers and oligomer radicals.³² At temperatures exceeding 500 °C, all of the polymers are completely decomposed without residual carbon. The DSC curves of the polymers show a step toward heat absorption, as evident in Figure 3b, and the temperature corresponding to the midpoint of this step is the glass transition temperature (T_g) of the polymers. The T_g is the lowest temperature at which a molecular chain segment can move, and its level is directly related to the flexibility of the molecular chain. It can be identified that the T_g of the PEA, PEAf, PEAS, PEASf, and PEASBF were 30 °C, 25 °C, 25 °C, 28 °C, and 38 °C, respectively. The results of TGA and DSC curves suggest that the IBOMA monomers can enhance the hardness of amphiphilic polymers, thereby enhancing their durability.

Figure 4 shows the adhesion strength and the water absorption of various polymers. Before immersion, the adhesion strength of PEA, PEAf, PEAS, PEASf, and PEASBF is 1.4, 1.2, 0.9, 0.8, and 1.0 MPa, respectively. All polymers are anchored primarily through polar interactions between polar acrylate groups in polymers and polar groups in epoxy board.³³ The total number of polar acrylic groups is approximately the same for different polymers, while the adhesion strength of PEA is higher than that of the other polymers. This phenomenon is attributed to the hydrophilic sulfobetaine groups and the low surface energy fluorinated groups that reduce the number of anchored polar acrylic groups, resulting in the reduction of adhesion strength.³⁰ After immersion, the

adhesion strength of all polymers decreased due to the permeation of water vapor into the polymers. The water absorptions of PEA, PEAf, PEAS, PEASf, and PEASBF are 3.8%, 3.5%, 9.4%, 7.6%, and 5.8%, respectively. The PEAS exhibits the highest water absorption among all these polymers due to the strong hydration capacities of zwitterionic groups through ionic solvation.¹⁸ Compared to PEAS, PEASBF exhibits a lower water absorption, which is attributed to the hydrophobicity of fluorination/isobornyl groups. Notably, PEASBF shows the least decrease in adhesion strength when compared to other polymers after water immersion, which is primarily attributed to the fact that the three-dimensional hydrophobic structure of IBOMA monomers serves to enhance the adhesion strength of amphiphilic polymers to substrates by reducing the internal stress of coatings.^{34,35} Subsequently, in a marine field test, even after four months of seawater immersion, none of the coatings fall off from the epoxy plates, indicating that the prepared coatings have a good stability in seawater (Figure S5, Supporting Information).

3.2. Surface Structure of the PEASBF. The antifouling performance of polymers is closely related to the surface morphology, surface wettability, and surface structure. The typical surface morphology and corresponding roughness of the as-prepared polymers are shown in Figure 5. Before immersion, all coatings maintained a smooth surface without wrinkles or cracks (S_a : ~ 10 nm), proving that PEASBF has good film-forming properties. After immersion in the artificial seawater for 7 days, only the surface of PEA and PEAf remains smooth without noticeable changes, which can be attributed to the absence of hydrophilic groups in its molecular chains. PEAS, PEASf and PEASBF coatings show microcreases on their surface after immersion, but the overall integrity of all the coatings remains intact with no cracks. The hydrophilic sulfobetaine groups inside the coating tend to move toward the surface of the coatings to contact water molecules. Such a movement of sulfobetaine groups will change the original molecular chain arrangement on the surface of the coating, affecting the surface morphology of the coatings and making the coating appear micro folds. Compared to PEAS, PEASf has a rougher surface after immersion (S_a (PEAS): 96 nm, S_a (PEASf): 790 nm), which can be ascribed to the incompatibility of hydrophobic fluorine monomers and hydrophilic sulfobetaine monomers. While the surface morphology of PEASBF becomes smoother than that of PEASf (S_a (PEASBF): 227 nm), which is attributed to the fact

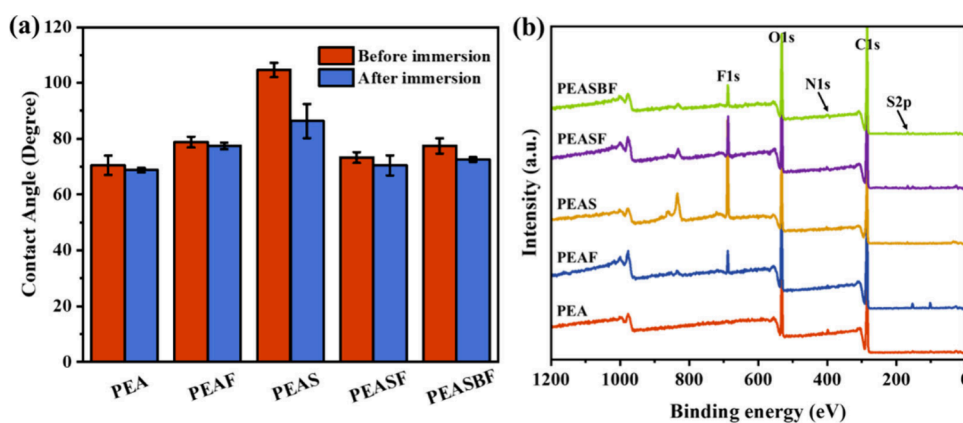


Figure 6. (a) Water contact angle of polymers before and after immersion. (b) XPS spectra of polymers after the triple purification process.

that the polarity of IBOMA monomers is intermediate between fluorine and sulfobetaine monomers, mitigating phase separation. More importantly, IBOMA monomers can supply cohesive strength, which ensures that there is less susceptibility to damage when polymers are in phase separation.^{8,12,21,34}

The adhesion processes of fouling proteins are related to the surface wettability and chemical structure of coatings.³⁶ The water contact angle of the coatings is shown in Figure 6a. It can be seen that the water contact angles of PEA and PEAF are about 70° and 80°, respectively. Surprisingly, the water contact angle of PEAS is as high as 105° before immersion, which is much higher than those of PEASF (73°) and PEASBF (77°). The water contact angle of zwitterionic polymers normally does not exceed 90°. Herein, the abnormally high contact angle of PEAS may be related to the 2,2,2-trifluoroethanol solvent, because the 2,2,2-trifluoroethanol is proton donating solvent with hydrophobic trifluoromethyl groups, which can form hydrogen bonding with N or O atoms in sulfobetaine structures.^{38–40} It is worth noting that the 2,2,2-trifluoroethanol is not completely removed by acetone dissolution, precipitation in deionized water, and finally vacuum drying at 80 °C, which can be further demonstrated by the XPS results that a significant amount of elemental fluorine remains present on the PEAS surface, as shown in Figure 6b. However, the 2,2,2-trifluoroethanol on the surface of polymers can release by immersion in water. As shown in Figure S2, the elemental fluorine of PEAS disappears from the surface after immersion, while it remains on the surface of PEASBF, imploring that copolymerized 2,2,2-trifluoroethyl methacrylate monomers are more stable on the polymer surface compared to 2,2,2-trifluoroethanol solvents. Besides, released 2,2,2-trifluoroethanol has no effect on biological activity due to the very low levels in the polymer network (Figure S3, Supporting Information). Accordingly, the contact angle of PEAS decreases significantly to 88° after immersion, precisely due to the migration of hydrophilic amphoteric ions toward the surface and the release of hydrophobic TFE into water. Additionally, the contact angle of PEASF and PEASBF is lower than that of PEAS before and after immersion in water. This was likely due to sulfobetaine monomers being copolymerized with fluorine monomers, which reduces binding energy between polymers and 2,2,2-trifluoroethanol and helps to alleviate the hydrogen bonding with 2,2,2-trifluoroethanol (Table S2, Supporting Information). Typically, the hydrophilic surfaces have better antiprotein adhesion properties than hydrophobic surfaces,³⁶ thus the exposure of more hydrophilic

components to the hydrophobic polymer surface can provide enhanced antiprotein adhesion properties of coatings.⁷

The chemical composition of the polymer surfaces is analyzed by XPS, as shown in Figure 6b. The C 1s and O 1s on the PEA surface are derived from MMA and EA monomers. The C 1s, O 1s, S 2p, N 1s, and F 1s are observed on the PEAS, PEASF, and PEASBF surfaces. The surface elemental concentration is summarized in Table 2. The PEAS surface

Table 2. XPS Atomic Surface Percentages of Polymers

Sample	Surface elemental concentration (atomic%)				
	C	O	S	N	F
PEA	79.19	20.81	-	-	-
PEAF	71.50	25.55	-	-	2.95
PEAS	61.11	20.68	0.50	0.93	16.78
PEASF	68.53	24.75	0.58	1.19	4.95
PEASBF	73.37	21.02	0.60	1.71	3.30

contains a significant content of the F element, accounting for 16.78%, which primarily originated from the 2,2,2-trifluoroethanol. After the copolymerization of sulfobetaine monomers with fluorinated/isobornyl monomers, the F element is only 3.30% on the PEASBF surface, which may be because sulfobetaine monomers are copolymerized with fluorine monomers to alleviate the hydrogen bonding with 2,2,2-trifluoroethanol.³⁸ It is noteworthy that the N and S elements increase progressively with a gradual participation of fluorine monomers and isobornyl monomers in the copolymerization process. This phenomenon suggests that the copolymerization of fluorinated or isobornyl monomers with sulfobetaine monomers facilitates the exposure of sulfobetaine groups on coating surfaces.

In addition, the XPS fine spectra of PEAS and PEASBF are compared to further reveal the effect of introducing fluorinated/isobornyl monomers into sulfobetaine polymers on the surface structure of polymers, as shown in Figure 7. Among them, the O 1s spectra and the S 1s spectra are not significantly different. The interaction of sulfobetaine groups with fluorinated/isobornyl groups or 2,2,2-trifluoroethanol mainly occurs on the C 1s spectra, the N 1s spectra, and the F 1s spectra. In the high-resolution F 1s spectra, the C–F from the TFEMA monomers in PEASBF is at 688.4 eV, whereas the –CF₂ binding energy from the TFE solvent in the PEAS is at 688.7 eV.^{21,41} The high-resolution C 1s spectra show that the –CF₃ peak only exists in PEAS, which is mainly because the

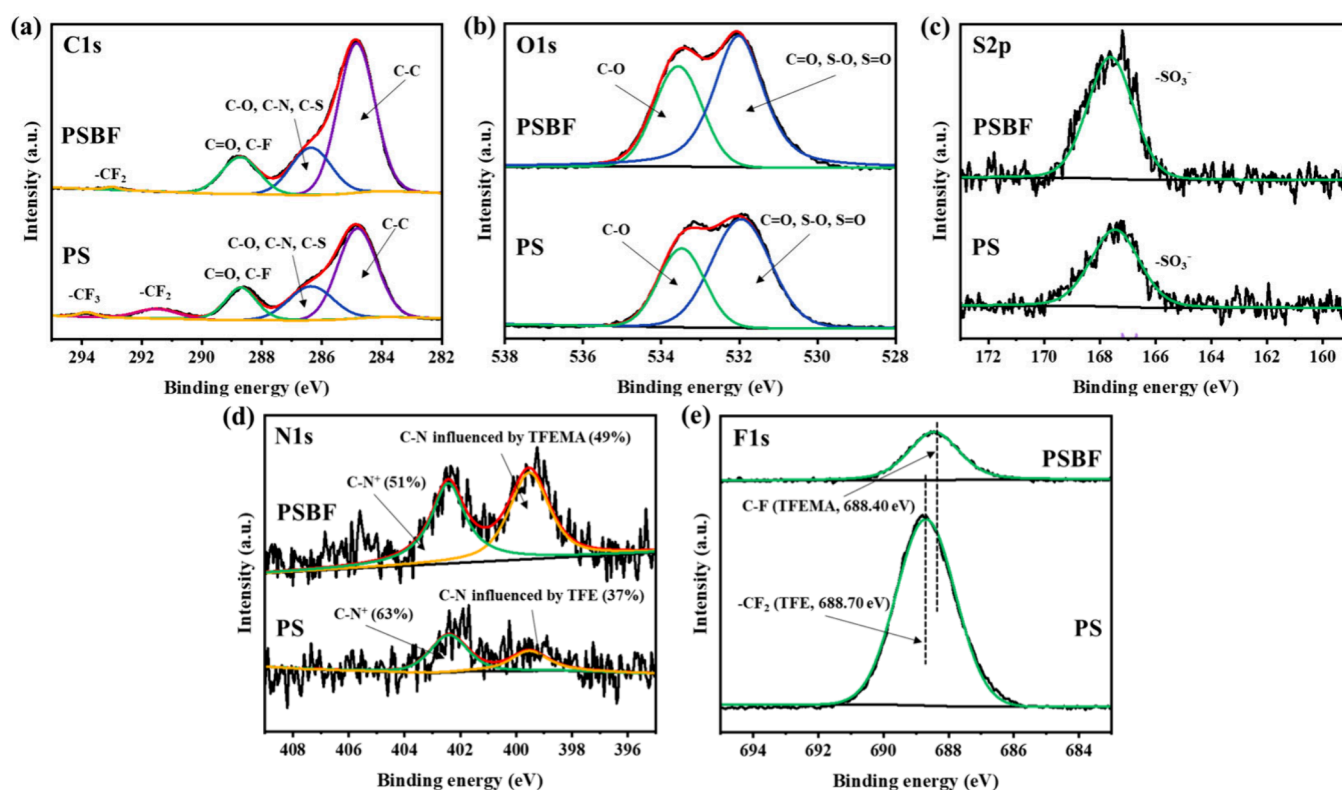


Figure 7. XPS fine spectra of PEAS and PEASBF: (a) C 1s, (b) O 1s, (c) S 2p, (d) N 1s, and (e) F 1s.

Table 3. Atomic Ratios for Polymers in Theory, Dry Conditions, and Wet Conditions^a

sample	O/C ratios in theory	O/C ratios in dry conditions			O/C ratios in wet conditions		
		0–10 Å	10–20 Å	>20 Å	0–10 Å	10–20 Å	>20 Å
PEA	0.400	0.391	0.405	0.404	0.416	0.396	0.399
PEAF	0.380	0.373	0.385	0.378	0.375	0.382	0.380
PEAS	0.417	0.438	0.415	0.414	0.396	0.428	0.426
PEASF	0.399	0.388	0.394	0.407	0.400	0.388	0.413
PEASBF	0.239	0.203	0.239	0.252	0.242	0.229	0.271

^aThe 0–10 Å, 10–20 Å, > 20 Å are defined as the polymer primer layer, intermediate layer, and top layer, respectively.

2,2,2-trifluoroethanol migrates to the surface more easily as molecular and fluorinated groups are easily limited by copolymerization chain segments during the migration process. Even so, copolymerized fluorinated groups are more likely to bring sulfobetaine groups to the surface during film formation compared to 2,2,2-trifluoroethanol. Notably, the high-resolution N 1s spectra enables the distinction between C–N⁺ and fluorine-induced C–N peaks observed at 402.4 and 399.6 eV, respectively.⁴² The ratio of integrated peak areas for fluorine-induced C–N and C–N⁺ peaks in PEASBF is higher than that in PEAS, suggesting that copolymerizing fluorinated groups is able to induce more sulfobetaine groups to surface exposure.⁴² This hydrophobic group induces more hydrophilic amphiphilic groups to be exposed to the surface, which may have a positive effect on the antifouling and fouling release properties of the coating. In addition, to prevent TFE from interfering with the antifouling and fouling release properties of PEAS, the original PEAS is heat-treated at high-temperature vacuum to remove residual TFE in polymers (Figure S6, Figure S7a–b), and such heat treatment does not damage the structure of polymers (Figure S7c–d). Detailed analysis of instructions for removing TFE from PEAS samples and the effect of TFE on antifouling

and fouling release properties of polymers can be found in the Supporting Information (Table S4, Figure S8).

3.3. MD Simulations of the PEASBF. The MD simulation is a very useful method to explore molecular mobility and structural surface properties at the atomic level, and this method is helpful to understand the antifouling mechanism of the polymers.^{43–45} To further explore how hydrophobic functional groups (TFEMA, IBOMA) affect hydrophilic functional groups (SBMA) of polymer surfaces, the atomic numbers of C and O are calculated along the z-axis. The calculating result of O/C atomic ratios for polymers in theory, dry conditions, and wet conditions is shown in Table 3. The O/C ratio of PEA and PEAS surface layer (>20 Å) are almost the same as the theoretical value in dry conditions, suggesting that MMA, EA, and SBMA monomers are evenly distributed in the surface layer during film formation. In theory, the O/C ratio of MMA, EA, SBMA, IBOMA, and TFEMA are 0.40, 0.40, 0.45, 0.14, and 0.33, respectively. SBMA monomers increase the O/C ratio in the system, while IBOMA and TFEMA monomers decrease the O/C ratio in the system. Therefore, a higher O/C ratio means more SBMA monomers in the layer. Compared to PEAS, the O/C ratios of

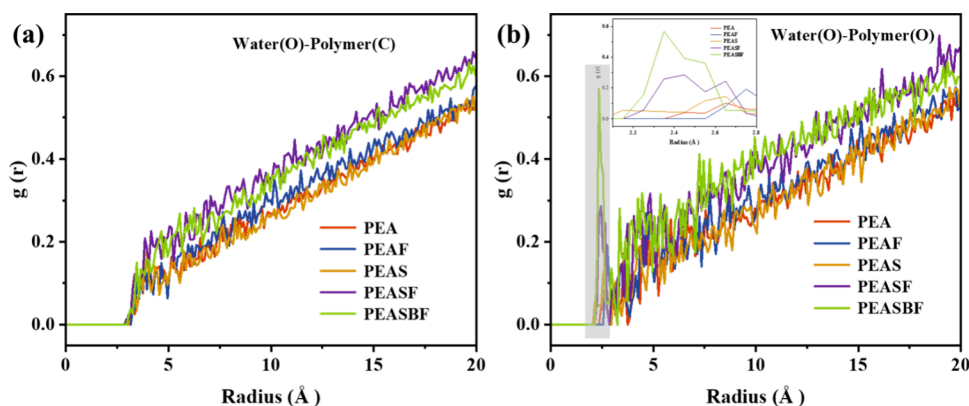


Figure 8. Radial distribution functions for the O of the water part with (a) C of polymers and (b) the O of polymers in wet condition.

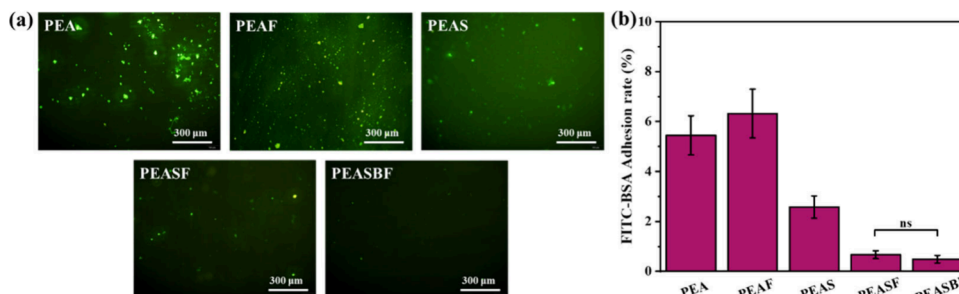


Figure 9. (a) Fluorescence microscopy images and (b) FITC-BSA adhesion rate of polymers after immersion in the FITC-BSA solution for 24 h. The value bars with “ns” not significant, asterisk denotes statistically significant differences, * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$.

PEASF and PEASBF surface layer (>20 Å) are higher than that of intermediate layer (10 – 20 Å), which proves that the O atoms of sulfobetaine groups from SBMA monomers driven by hydrophobic TFEMA and IBOMA monomers are more readily exposed to coating surfaces. This conclusion is consistent with the result of the XPS atomic surface percentages. On the other hand, in the presence of water, the O/C ratios of PEAS, PEASF and PEASBF surface layer (>20 Å) in wet conditions are higher than that in dry conditions, attributed to the migrate tendency of hydrophilic sulfobetaine groups into water.

Radial distribution function (RDF) is commonly used to characterize extended molecular structure, which represents the probability of finding a pair of atoms at a distance r (radius) with respect to the bulk in a completely random distribution.^{43,44} The interactions between the water oxygen atoms toward carbon and oxygen atoms of the polymeric layer are explored by RDF analysis, as shown in Figure 8. Whether in Water(O)-Polymer(C) or in Water(O)-Polymer(O) RDF analysis, the $g(r)$ values of PEASF and PEASBF are higher than those of PEA, PEAF, and PEAS, proving that the large number of the water molecules are located at the vicinity of the top layer surface of PEASF and PEASBF. Noteworthy, the RDF between oxygen of polymers with the water oxygen atoms has a peak in the range of 2.2 to 2.8 Å, as shown in Figure 8b, which corresponds to the formation of ionic solvation from SBMA monomers. This indicates that the SBMA monomers of polymers have a stronger interaction with water molecules on the surface layer. Similar results can be derived from the binding energy data (Table S3 in the Supporting Information). PEASBF has the highest peak value and hydration energies (-744.3 ± 4.7 kJ/mol), indicating stronger interactions, while PEAS with the same amount of SBMA monomers as PEASBF

has weaker peak value and hydration energies (-347.7 ± 9.3 kJ/mol). That is, even if the amount of hydrophilic SBMA monomers within the polymer is the same, interaction strength is enhanced by the migration of SBMA from the intermediate layer to the surface layer. Consequently, PEASBF exhibits a better performance according to its tendency to adsorb much more water molecules to its surfaces.

3.4. Antifouling and Fouling Release Properties of the PEASBF. The antiprotein adhesion properties are important for antifouling coatings because the initial process of biofouling formation during the conditioned film formation stage involves a protein adsorption process, and some fouling organisms secrete protein-based adhesives to help themselves settle on underwater surfaces.¹⁹ Bovine serum albumin (BSA), a globulin in bovine serum, is commonly used to evaluate the antiprotein adhesion properties of materials, and the testing results are shown in Figure 9. It is observed that large amounts of proteins adhere on the surface of PEA (5.5%), PEAF (6.3%), and PEAS (2.6%), whereas only small amounts of proteins adhere on the surface of PEASF (0.7%) and PEASBF (0.5%). Compared to PEA and PEAF, the PEAS has a better antiprotein adhesion property. This observation may be ascribed to the fact that the hydrophilic sulfobetaine groups can weaken the protein adhesion on hydrophobic surfaces. Furthermore, the antiprotein adhesion property of PEASF and PEASBF is superior to that of PEAS. This is because the hydrophobic fluorinated/isobornyl groups induces more sulfobetaine groups to migrate toward the surface during the film formation process, resulting in an amphiphilic surface. The amphiphilic surface with more hydrophilic sulfobetaine groups can adsorb much more water molecules to its surfaces, which can effectively resist nonspecific hydrophobic interactions between protein and polymer surfaces.^{28,36}

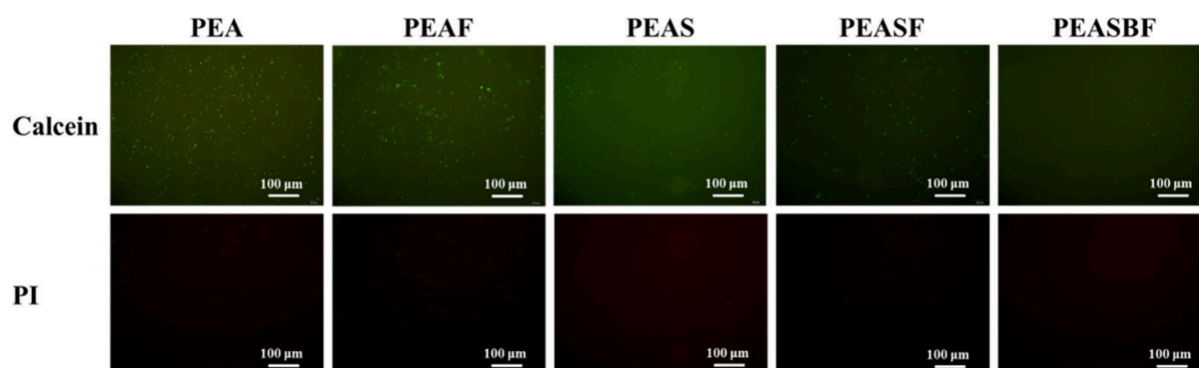


Figure 10. Fluorescence photographs of *Pseudomonas sp.* adhered on the surface of polymers — green color indicates alive bacteria (Calcein) and red color indicates dead bacteria (PI).

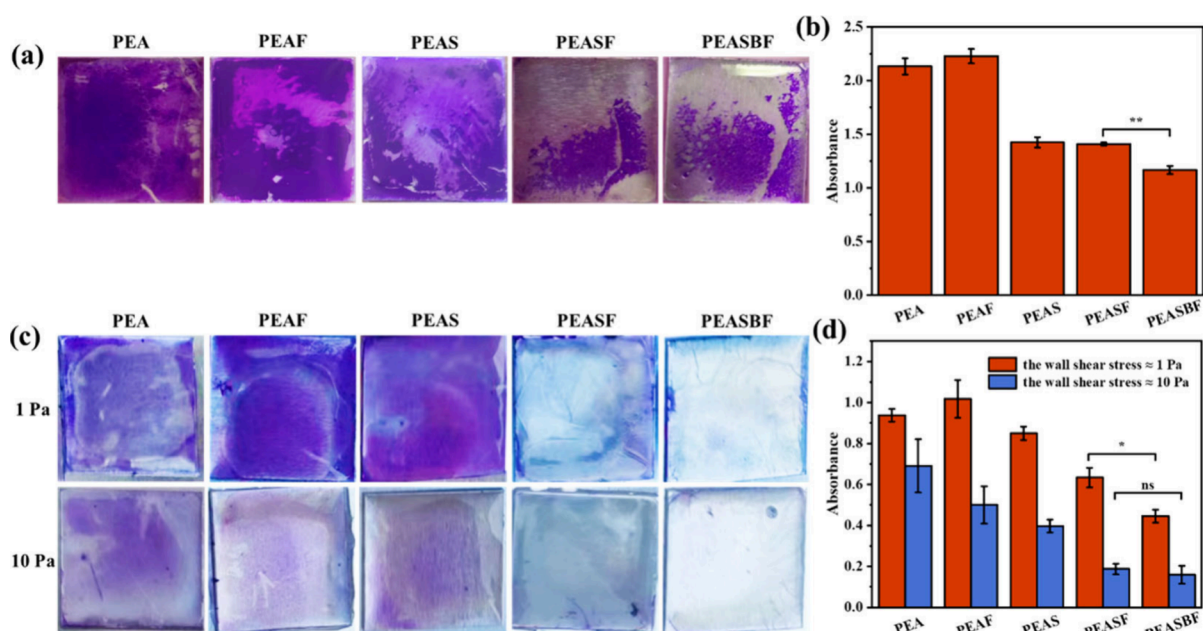


Figure 11. (a) The attachment biofilm images of *Pseudomonas sp.* (a) and (b) the corresponding absorbance of crystal violet after 0.2 Pa shear stress for 1 min. (c) The attachment biofilm images of *Pseudomonas sp.* and (d) the corresponding absorbance of crystal violet after 1 Pa, 10 Pa shear stress for 1 min. The value bars with “ns” not significant, asterisk denotes statistically significant differences, * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$.

Pseudomonas sp. readily forms biofilms on material surfaces and is used to evaluate the antimicrobial activity of coating. The growth curves of *Pseudomonas sp.* in medium with different polymers are shown in Figure S3. It can be seen that the growth curves of the PEA, PEAf, PEAS, PEASf, and PEASBF are the same as that of the blank, indicating that all coatings cannot inhibit the growth of bacteria. The bactericidal properties of coating surfaces are further investigated by dead-viable surface staining, as shown in Figure 10. The alive bacteria are colored green by the Calcein dye, and the dead bacteria are colored red by the PI dye. Observations reveal the presence of numerous green bacteria and the absence of dead red bacteria on the surfaces of PEA, PEAf, and PEAS, demonstrating that their surfaces do not possess bactericidal activity. The PEASf surface exhibits a small number of green bacteria and no red dead bacteria, suggesting that the amphiphilic surfaces, composed of hydrophobic fluorinated groups and hydrophilic sulfobetaine groups, can prevent bacterial adhesion effectively. Additionally, fewer green bacteria adhere on the PEASBF surface compared with the

PEASf, indicating that the isobornyl ester structure enhances the antimicrobial activity of amphiphilic polymers. The isobornyl ester structure may resist bacterial adherence through a specialized icosahedral structure rather than directly killing the bacteria.⁴⁶

Subsequently, the biofilm resistance of polymers by crystal violet staining is evaluated, and the formed *Pseudomonas sp.* biofilms are stained with crystal violet, and the absorbance value is proportional to the number of biofilms on the coating, as shown in Figure 11a–b. The entire surface of PEA is stained purple due to the presence of crystalline violet, and the absorbance of PEA is as high as 2.1, which indicates that PEA without functional groups cannot inhibit the formation of a biofilm. The PEAf with fluorinated groups is also stained purple on its surface, indicating that the fluorinated groups cannot inhibit bacterial adhesion. In addition, the PEAS surface and the PEASf surface are partially covered by crystalline violet, and their absorbance is approximately 1.4, suggesting that the hydrophilic sulfobetaine groups can inhibit the formation of biofilm. This suggests that the ability of

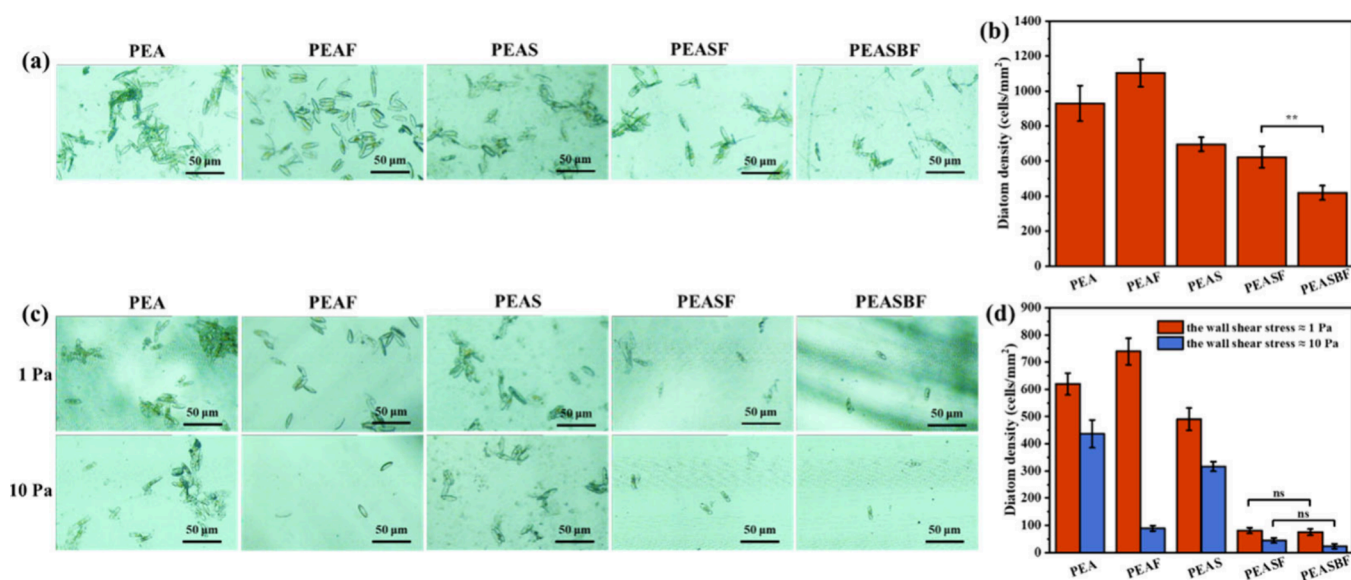


Figure 12. (a) The attachment diatom images and (b) diatom density of *Navicula sp.* after 0.2 Pa shear stress for 1 min. (c) The attachment diatom images and (d) diatom density of *Navicula sp.* after 1 and 10 Pa shear stress for 1 min. The value bars with “ns” not significant, asterisk denotes statistically significant differences, * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$.

amphiphilic polymers to resist bacterial biofilms derives mainly from the contribution of hydrophilic sulfobetaine groups. The area of crystalline violet on the surface of PEASBF is lower than that of PEASF, and the corresponding absorbance is 1.1, suggesting that the steric structure of isobornyl groups can enhance the antibiofilm formation of amphiphilic polymers. Then, the biofilm removal test is performed at the shear pressure of 1 and 10 Pa, respectively, as shown in Figure 11c-d. The PEA surface still has a large area of crystalline violet whether at a shear pressure of 1 or 10 Pa, and the absorbance is as high as 0.9 under a shear pressure of 1 Pa, indicating that the attached biofilm is difficult to remove from the PEA surface. The PEAf surface has a smaller amount of crystalline violet at a shear pressure of 10 Pa compared with PEA, suggesting that fluorinated groups promote bacterial release from surfaces under high shear pressure. The biofilm removal ability of PEASF is superior to that of PEAS both at low (1 Pa) and high (10 Pa) shear pressure, which may be because copolymerized 2,2,2-trifluoroethyl methacrylate monomers readily form an amphiphilic surface with sulfobetaine groups. The amphiphilic surface consisting of hydrophobic fluorinated groups and hydrophilic sulfobetaine groups is more likely to release adherent bacteria than surfaces modified by a single functional group. For PEASBF, whether under low or high shear pressure, the crystalline violet region on the PEASBF surface is absent and the absorbance is only 0.4, suggesting that the three-dimensional structure of isobornyl groups on the amphiphilic polymer surface can effectively enhance the biofilm removal ability. Noteworthy, at high shear pressures (10 Pa), no significant purple areas are observed on PEASF and PEASBF surfaces, and their absorbance is about 0.2. At low shear pressures (1 Pa), the PEASBF surface consisted of hydrophobic fluorinated structures, hydrophilic sulfobetaine zwitterion structures, and antimicrobial isobornyl structures, which exhibits the strongest biofilm removal capacity. In summary, PEASBF exhibits properties of both inhibiting the formation of bacterial biofilms on the surface and releasing already formed bacterial biofilms due to its antimicrobial amphiphilic surfaces.

Navicula sp., as typical fouling organisms, is also a food source for macrofouling organisms. Therefore, the study of diatom adhesion and release is important to investigate the performance of antifouling coatings. In previous studies, wall shear stresses between 0.1 and 0.3 Pa are found to be suitable for distinguishing cell attachment densities on different surface chemistries.^{29,47} Wall shear stresses above 0.5 Pa are found to be suitable for studying cell release from different surface chemistries.^{47,48} Here, the antidiatom adhesion properties of coatings are tested at 0.2 Pa and the diatom release properties of coatings are assessed at 1 and 10 Pa. The results of the antidiatom adhesion test and the diatom release test are shown in Figure 12. Both at low (0.2 Pa) and high (1 Pa, 10 Pa) shear pressures, large amounts of diatoms are observed on the surface of PEA, indicating that the antidiatom adhesion and diatom desorption performance of PEA is abysmal. Whether at 0.2 or 1 Pa shear pressure, many diatoms adhere to the surface of PEAf, while only a small number of diatoms adhere to the surface of PEAf at 10 Pa shear pressure, which indicates that fluorinated groups do not inhibit diatom adhesion but promote diatom release. Unlike PEAf, the PEAS has a better resistance to diatom adhesion at shear pressures of 0.2 and 1 Pa, while many diatoms still adhere to the surface at 10 Pa shear pressure, suggesting that sulfobetaine groups mainly play a role in preventing diatom adhesion. Due to the combination of hydrophilic sulfobetaine groups and hydrophobic fluorinated groups, the PEASF exhibits superior resistance to diatom adhesion and diatom release properties compared to PEAf and PEAS. Only a very small amount of diatoms is observed on the surface of PEASBF at a shear pressure of 0.2 Pa, and almost no diatoms are observed on the surface of PEASBF at a shear pressure of 10 Pa. Figure S4 further shows the diatom attachment rate and diatom removal rate of the coatings. Compared with other polymers, PEASBF has the lowest diatom adhesion rate of 33% at 0.2 Pa and the highest diatom desorption rate of 98% at 10 Pa. This excellent performance is because PEASBF combines three functional groups for antifouling, including the SBMA, TFEMA, and IBOMA component. Among them, the hydrophilic SBMA component

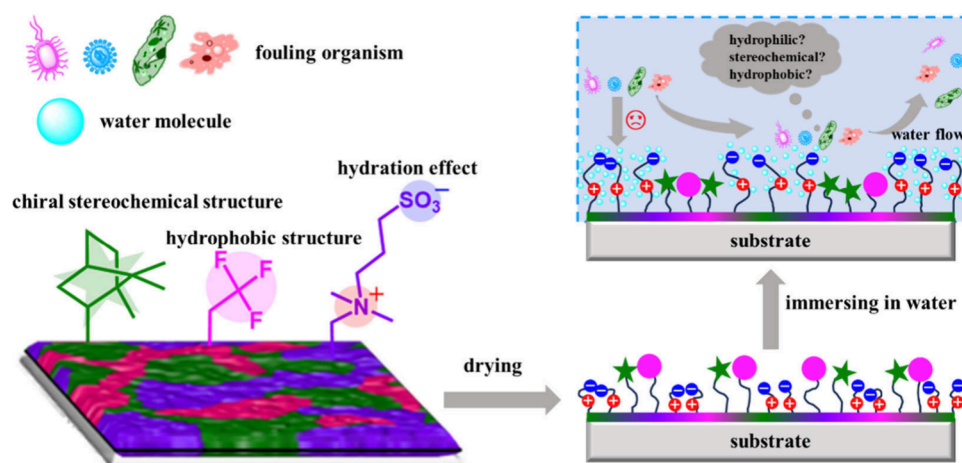


Figure 13. Antifouling schematic diagram of the PEASBF coating.

inhibits the initial adhesion of marine diatoms, while the hydrophobic TFEMA component weakens their adhesion to the solid surface. At the same time, the antibacterial IBOMA component reduces the density of bacterial biofilm, which is helpful in inhibiting the attachment of diatoms.^{21,42} Consequently, PEASBF has excellent resistance to diatom adhesion and diatom release properties.

The adhesion of typical biologicals on the PEASBF surface, like proteins, bacteria, and diatoms, is also studied to prove that PEASBF not only can inhibit the attachment of typical biofouling organisms but also is prone to release adherent biofouling organisms. The antifouling schematic diagram of PEASBF is shown in Figure 13. Initially, the chiral stereochemical structure of IBOMA, hydrophilic structure of SBMA, and hydrophobic structure of TFEMA together form the surface of PEASBF, and the SBMA structure will migrate more to the surface driven by the TFEMA and IBOMA structure during drying. Subsequently, when PEASBF is immersed in the seawater, the proteins first adheres to the PEASBF surface through noncovalent interactions. At this point, the sulfobetaine groups in the SBMA component of the polymer increases the hydrophilicity of the coating surface, which reduces the hydrophobicity of the proteins with the surface and mitigates the adhesion of proteins to the surface.⁴² Then bacteria from the marine environment started to attack the coated surface. At this point, the sulfobetaine groups in the SBMA component and the fluorinated groups in the TFEMA component work together to form an amphiphilic surface, reducing physicochemical interactions between bacteria and the PEASBF surface to inhibit primary attachment.^{3,42} At the same time, the chiral stereochemical isobornyl groups in the IBOMA component affects the surface recognition of bacteria on the surface and reduces the density of bacterial biofilms.^{21,26,46} When diatoms attack the PSBF surface, there is no primary fouling biofilm formed on the surface, which is not conducive to adhesion. Furthermore, the hydrophobic TFEMA structure, the chiral stereochemical IBOMA structure, and the hydrophilic SBMA structure can commonly construct an “ambiguous” surface, which can “confuse” organisms during settlement and adhesion to reduce the bonding between the attaching diatoms and PEASBF surfaces, facilitating the release of the adhered diatoms from surfaces under low water flow shear stress.^{8,21,42} Other than proteins, bacteria, and diatoms, there are many organisms in the ocean, e.g., algae, seagrasses,

invertebrates, etc. The formed microfouling film consisting of proteins, bacteria, and diatoms stimulates the recruitment of animal larvae, such as barnacles and mussels, which belong to macrofouling. Here, the PEASBF can reduce the settlement of animal larvae by inhibiting the formation of microfouling film in the marine (Figure S5, Supporting Information). Therefore, the antifouling properties and release properties of the PEASBF coating are achieved through the combination of the excellent antimicrobial properties of IBOMA components, the low surface energy properties of TFEMA components, and the hydrophilic properties of SBMA components, making them environmentally friendly antifouling coatings with excellent antifouling effects and fouling release properties.

4. CONCLUSIONS

In summary, a new sulfobetaine borneol fluorinated polymer was successfully synthesized through a free radical polymerization process, utilizing 2,2,2-trifluoroethanol and 2-acetoxy-1-methoxypropane as solvents. The XPS results and MD simulations revealed that the copolymerizing of fluorine and isobornyl could induce a higher exposure of sulfobetaine on the surfaces of PEASBF (N elemental concentration: 1.71%) compared to PEAS (N elemental concentration: 0.93%). The amphiphilic PEASBF coating with the chiral stereochemical structure of borneol exhibited outstanding antifouling and fouling release properties. Specifically, the PEASBF could resist the adhesion of *Bovine serum albumin* (adhesion rate: 0.5%), *Pseudomonas sp.* (Biofilm: 1.3 absorbance), and *Navicula sp.* (Diatom attachment rate: 33%). Furthermore, the PEASBF could effectively remove adhered *Pseudomonas sp.* and *Navicula sp.* (Diatom removal rate: 33%) under a shear stress of 10 Pa. Therefore, this study holds promise for the development of nontoxic marine antifouling coatings that are suitable for static applications.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acsami.4c06525>.

Additional ATR-FTIR, ¹H NMR, MD simulation, XPS spectra results; instructions for removing TFE from PEAS samples; the effect of TFE on antifouling and fouling release properties of polymers; marine field test (PDF)

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Notes

The authors declare no competing financial interest.

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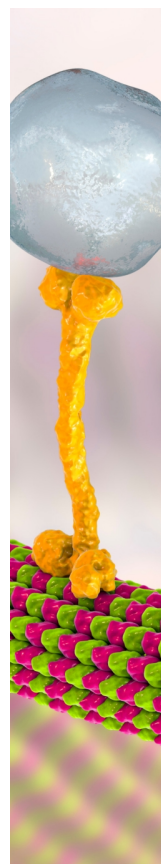
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